HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Java Spring Boot** REST APIs for internal service communication, processing approximately **1,500 requests per minute** during peak hours.
- Developed and maintained **Hibernate** entities and queries for a core data persistence layer, ensuring data integrity across multiple enterprise applications.
- Wrote comprehensive **JUnit** and **Mockito** test suites for critical Java modules, increasing code coverage by **22%** and reducing post-deployment defects.
- Analyzed and modelled fire growth, structural fire resistance, and smoke control scenarios for new and existing building projects using **Computational Fluid Dynamics** software.
- Prepared detailed technical reports and presentations on fire engineering analysis findings for clients, including major property developers and construction firms.
- Collaborated with senior engineers to develop innovative fire protection strategies for diverse building types, such as high-rise offices and large retail complexes.
- Containerised multiple backend services using **Docker** and deployed them to a managed **AWS EKS** cluster, streamlining the deployment pipeline.

SKILLS

Core Engineering – Fire Engineering Analysis, Computational Fluid Dynamics, Structural Fire Resistance, Smoke Control, Means of Escape

Programming Languages – Java, Python, SQL, English

Frameworks & Libraries – Spring Boot, Hibernate, JUnit, Mockito, pandas, NumPy, scikit-learn

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Agile, Scrum, Jira, Confluence, Splunk, Dynatrace, Power BI, Tableau

PROJECTS

Enterprise Java Application Modernisation

- Refactored monolithic Java application components into microservices using **Spring Boot** and **Hibernate**, improving system modularity and maintainability.
- Developed RESTful APIs to facilitate inter-service communication, handling approximately **500,000 transactions daily** and reducing interdependencies.
- Implemented robust error handling and logging mechanisms across services, contributing to a **15% reduction** in critical production incidents.

Building Fire Safety Analysis Tool

- Developed a Python-based simulation tool to assist in analyzing fire spread and escape routes for various building structures, integrating basic physics models.
- Designed and implemented data ingestion pipelines using **pandas** to process building schematics and material properties for simulation input.
- Created data visualizations with **Matplotlib** to represent simulation outputs, aiding engineers in understanding potential fire scenarios and escape path efficiency.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)